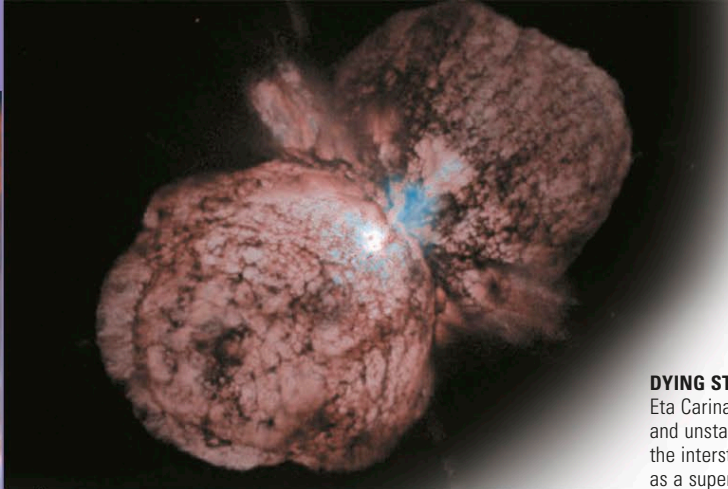




OLD STARS

Some of the most visible and familiar bodies in the sky are stars that are approaching the ends of their lives or are experiencing their final death throes. In Wolf-Rayet stars and planetary nebulae, these old stars also present some of the most dramatic events and most beautiful sights in the universe. Although different types of old stars exist throughout the Milky Way, the oldest are situated far out in the galactic halo (see pp.226–229) or within the globular clusters (see pp.288–295). Some of these stars are nearly as old as the universe itself.

DYING STAR

Eta Carinae is a large, extremely old, and unstable star ejecting material into the interstellar medium. It could explode as a supernova at any time.

RED GIANT

Aldebaran



TAURUS

DISTANCE FROM SUN
67 light-years
MAGNITUDE 0.85
SPECTRAL TYPE K5

Also known as Alpha (α) Tauri, Aldebaran is the brightest star in the constellation Taurus and the 13th-brightest star in the sky. Its surface temperature of only 6,740°F (3,730°C) makes it glow a dull red that can easily be seen by the naked eye. Aldebaran's diameter is about 45 times that of the Sun and, in place of the Sun, it would extend halfway to the orbit of Mercury. The star appears to be part of the Hyades cluster (see p.290), but this is a line-of-sight effect, with Aldebaran lying about 40 light-years closer to the Sun. This elderly star is a slow rotator, taking 2 years for each rotation, and an irregular variable, pulsating erratically. It has at least two faint stellar companions. Its name is derived from the Arabic *Al Dabaran*, meaning "the Follower," because it rises after



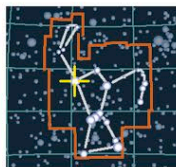
BULL'S EYE

The red tinge of Aldebaran makes it very distinctive against the whiter stars of the Hyades cluster. It is often depicted as the eye of the bull in the constellation Taurus.

the prominent Pleiades star cluster and pursues it across the sky. Aldebaran was one of the Royal Stars or Guardians of the Sky of ancient Persian astronomers and marked the coming of spring.

RED SUPERGIANT

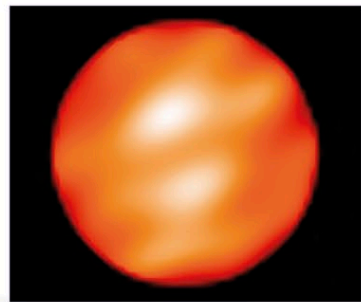
Betelgeuse



ORION

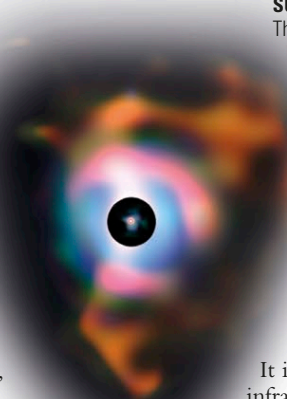
DISTANCE FROM SUN
500 light-years
MAGNITUDE 0.5
SPECTRAL TYPE M2

The right shoulder of the hunter, Orion, is marked by this distinctive, bright red star. Betelgeuse, or Alpha (α) Orionis, is a massive supergiant and the first star after the Sun to have its size reliably determined. Its diameter is more than twice that of the orbit of Mars, or about 500 times that of the Sun, and because of its huge size, it is about 14,000 times brighter. Betelgeuse is the 10th-brightest star in the sky, although as it pulsates, its brightness varies over a period of about 6 years.



SURFACE SPOTS

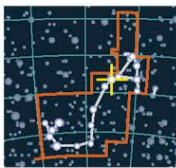
The infrared image of Betelgeuse above shows bright surface spots that could be convection cells. The infrared image at left shows gas and dust shed by the star, which has been masked by a black disk so that the gas and dust are visible.



It is a strong emitter of infrared radiation, which is produced by three concentric shells of material ejected by the star over its lifetime. It is slowly using up its remaining fuel and one day will probably explode as a supernova.

RED SUPERGIANT

Antares



SCORPIUS

DISTANCE FROM SUN
520 light-years
MAGNITUDE 0.96
SPECTRAL TYPE M1.5

Antares or Alpha (α) Scorpii is the 15th-brightest star in the sky. Estimates of its diameter range from 280 to 700 times that of the Sun. It is about 15 times more massive than the Sun and shines 10,000 times brighter. This elderly star pulsates irregularly and has a binary companion that orbits in a period of about 1,000 years. This companion lies close enough to be affected by Antares' stellar wind and is a hot radio source. When viewed through an optical telescope, this blue companion looks green because of the color contrast with red Antares.

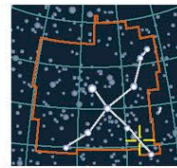
A RIVAL OF MARS

The glowing Antares (bottom right) looks a lot like Mars, the red planet. Its name derives from the Greek for "rival of Mars" (or *anti Ares*).



RED GIANT

TT Cygni



CYGNUS

DISTANCE FROM SUN
1,500 light-years
MAGNITUDE 7.55
SPECTRAL TYPE G

With a high ratio of carbon to oxygen in its surface layers, TT Cygni is known as a carbon star. The carbon, produced during helium burning, has been dredged up from inside the star. An outer shell, about half a light-year across, was emitted about 6,000 years before the star was as it appears to us now.

CARBON RING

This false-color image shows a shell of carbon monoxide surrounding the carbon star TT Cygni.

